

FastAPI Pagination Issue SOP

Objective

Resolve incorrect pagination behavior where page number, limit, offset, cursor, or total count does not match expected API behavior.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

- 1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
- 2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
- 3. Permission to view application configuration, environment variables, and deployment status.
- 4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
- 5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T050
Issue	FastAPI pagination issue
Category	API
Service	FastAPI
Severity	Low
Status	Resolved
Description	Pagination not working correctly
Expected Resolution	Fix query logic

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, API gateway logs, and monitoring dashboards to validate this incident in runtime.

Procedure

- 1. Confirm endpoint, pagination style, request parameters, and expected result.
- 2. Review actual response for page, size, offset, total count, next cursor, and item count.
- 3. Check query logic for off-by-one errors and default parameter handling.
- 4. Validate sorting order because pagination without stable ordering can duplicate or skip rows.
- 5. Check limit bounds and negative or missing parameter validation.
- 6. Compare API behavior against contract or frontend expectation.
- 7. Fix query logic, response metadata, and validation constraints.
- 8. Add tests for first page, last page, empty page, and boundary page sizes.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
```

```
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Wrong page results	Offset calculation error	Fix page-to-offset logic
Duplicate records	Unstable sorting	Add deterministic order by
Invalid limit accepted	Missing validation	Add query parameter constraints
Wrong total count	Incorrect count query	Fix count query and filters

Best Practices

- 1. Use stable sorting for all paginated endpoints.
- 2. Set maximum page size limits.
- 3. Test boundary cases.
- 4. Document pagination contract for frontend teams.

References

FastAPI Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | API Gateway Logs | Backend Runbook